Kummer lines of elliptic curves

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1. Group law on elliptic curves: adding points

$$E : y^2 = x^3 + ax + b \qquad \text{or} \qquad E_{\mathbb{P}} : Y^2Z = X^3 + aXZ^2 + bZ^3$$

- Projective coordinates of a point $(x, y) \in E$: $(X : Y : Z) \in E_{\mathbb{P}}$ where $x = \frac{X}{Z}$ and $y = \frac{Y}{Z}$, then $(X : Y : Z) = (\lambda X : \lambda Y : \lambda Z)$ for all λ.
- Point at infinity $\mathcal{O} = (0 : 1 : 0) \in E_{\mathbb{P}}$.

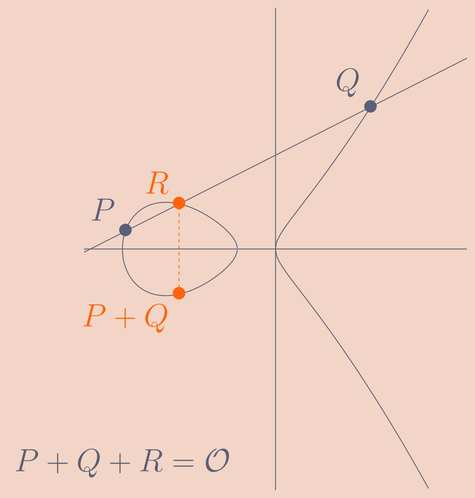

Fig. 1: Computing $P + Q$ with $P \neq Q$

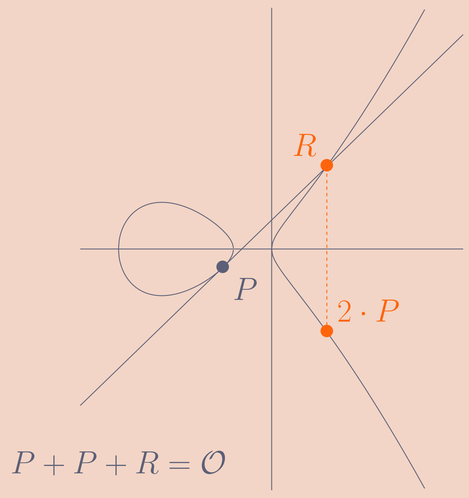

Fig. 2: Computing $2 \cdot P$

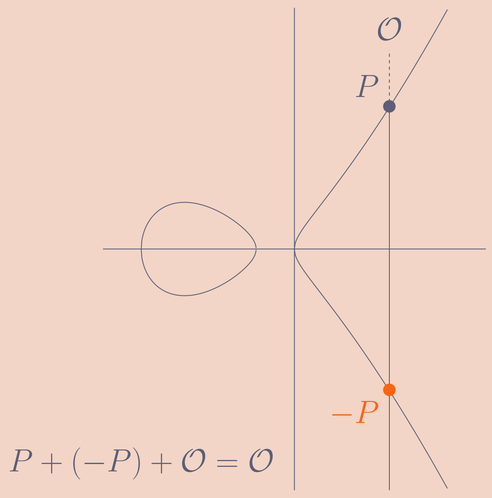

Fig. 3: Opposite of a point is the symmetric

In cryptography, it is crucial to compute $n \cdot P = P + \cdots + P$ efficiently.

2. Factoring integers: Elliptic Curve Method

Lenstra [1] introduced ECM in 1987. Say we want to factor an integer n. The algorithm is as follows:

1. Pick a random curve E with a random point $P = (x_0, y_0)$ on it: one can for instance choose $a, x_0, y_0 \in \mathbb{Z}/n\mathbb{Z}$ at random, and then set $b = y_0^2 - x_0^3 - ax_0$. Consider $E : y^2 = x^3 + ax + b$, then $P \in E$.
2. Choose a bound B not "too large", one wants to compute $B! \cdot P$. This can be done recursively by first computing $2 \cdot P$, then $3 \cdot (2 \cdot P)$ and so on... With the geometric law, this will involve computing slopes $\lambda = \frac{u}{v}$, and requires v to be invertible modulo n.
3. Two cases can occur:
   - If for every slope $\gcd(v, n) = 1$, the computation will end. We can then try with another curve.
   - If for some slope $\gcd(v, n) \neq 1$, then either $\gcd(v, n) = n$, in that case no information is gained, so we try with another curve, and otherwise $1 < d = \gcd(v, n) < n$ with $d \mid n$, so d is a non-trivial divisor.

The novelty of this approach is that it is really easy to pick another curve if the computation succeeds, or it is possible to restrict to curves with "good properties".

3. Different models of elliptic curves

- Short Weierstrass curve: $y^2 = x^3 + ax + b$.
- Legendre curve: $y^2 = x(x - 1)(x - \lambda)$.
- Montgomery curve: $By^2 = x(x^2 + \mathcal{A}x + 1)$.
- Twisted Edwards curve: $ax^2 + y^2 = 1 + dx^2y^2$.

- Addition of 2 points on short Weierstrass: $14\mathbf{M} + 1\mathbf{S} + 1\mathbf{m}_0$.
- Addition of 2 points on twisted Edwards: $10\mathbf{M} + 1\mathbf{S}$.

M: Multiplication / **S**: Square / $\mathbf{m}_0$: Multiplication by a constant

4. Kummer lines: is the y-coordinate essential?

If we know $x(P)$ the x-coordinate of a point $P \in E$, it is easy to recover $y(P)$ up to a sign with the equation. Can we do computations only with the x-coordinate?

From the equations, we also have that $x(P) = x(Q)$ if and only if $P = \pm Q$. We then introduce $\mathcal{K} = E/\pm 1$ and look at the map:

$$\begin{aligned} E &\to \mathcal{K} \simeq \mathbb{P}^1 \\ (x, y) &\mapsto (x : 1) = (X : Z) \\ \mathcal{O} &\mapsto \infty := (1 : 0) \end{aligned}$$

It is not a group any more! But we can still grab $x(P + Q)$ from $x(P)$, $x(Q)$ and $x(P - Q)$.

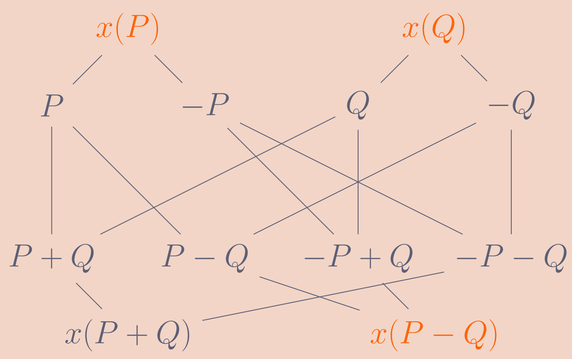

Fig. 4: Differential addition

5. Montgomery arithmetic

- Consider the elliptic curve $y^2 = x(x^2 + \mathcal{A}x + 1)$, Montgomery [2] gave the following formulas for x-only arithmetic:

$$\text{If } P \neq Q,\ x(P + Q)x(P - Q)(x(P) - x(Q))^2 = (x(P)x(Q) - 1)^2$$

$$\text{If } P = Q,\ 4x(2 \cdot P)x(P)(x(P)^2 + \mathcal{A}x(P) + 1) = (x(P)^2 - 1)^2$$

- On the curve $y^2 = x(x - \alpha)(x - \alpha^{-1})$, if $T = (\alpha, 0)$, then:

$$\alpha x(2 \cdot P + T)[2\alpha x(P) - (x(P)^2 + 1)]^2 = [\alpha(x(P)^2 + 1) - 2x(P)]^2$$

Complexities: $3\mathbf{M} + 2\mathbf{S}$ / $2\mathbf{M} + 2\mathbf{S} + 1\mathbf{m}_0$ / $4\mathbf{S} + 2\mathbf{m}_0$.

6. Computing $x(n \cdot P)$: to build a ladder

$$11 = 1 \times 2^3 + 0 \times 2^2 + 1 \times 2^1 + 1 \times 2^0$$

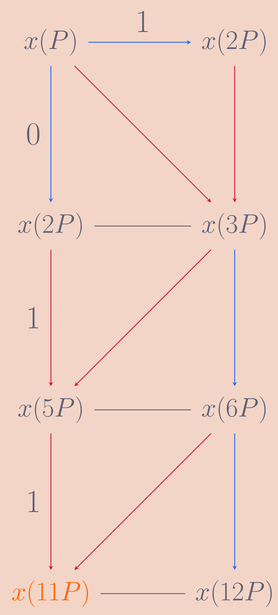

Fig. 5: Montgomery Ladder

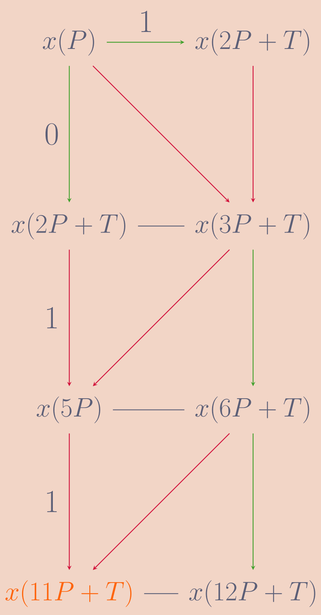

Fig. 6: Hybrid Ladder

References

[1] Hendrik W. jun. Lenstra. "Factoring integers with elliptic curves". English. In: *Annals of Mathematics. Second Series* 126 (1987), pp. 649–673. ISSN: 0003-486X. DOI: 10.2307/1971363.

[2] Peter L. Montgomery. "Speeding the Pollard and elliptic curve methods of factorization". English. In: *Mathematics of Computation* 48 (1987), pp. 243–264. ISSN: 0025-5718. DOI: 10.2307/2007888.

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